

Poster: MobiChem: A Ubiquitous Smartphone-Based Toolkit for Practical Fruit Monitoring and Analysis

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ABSTRACT

This paper introduces MobiChem, a low-cost, portable, practical, and ubiquitous smartphone-based toolkit for fruit monitoring. The key idea is to leverage the light emitted from a smartphone's screen and front camera, coupled with a custom-built screen cover, to perform comprehensive hyperspectral analysis on targeted objects. Specifically, we designed a zero-powered screen cover that selectively filters wavelengths essential for hyperspectral sensing. We then incorporate a CNN-based algorithm and a novel ranking-based learning technique that manipulates the latent space to classify maturity stages and characterize their chemical and physical factors. We showcased its application in tomato, banana, and avocado sensing. Our system examines the maturity, chlorophyll, lycopene content, free sugar levels, and firmness, enabling various dietary assessments and food safety applications.

CCS CONCEPTS

• Human-centered computing → Smartphones; Mobile phones.

KEYWORDS

Fruit Sensing, Hyperspectral Imaging, Mobile Sensing

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1 INTRODUCTION

Assessing fruit maturity is a multifaceted process that involves analyzing factors such as color, internal quality, and firmness. For example, tomatoes and avocados are rich in many vitamins and minerals and contain bioactive compounds such as carotenoids, flavonoids, and phenolic acids, which contribute to their antioxidant properties [3]. However, as the fruits ripen, they undergo significant physiological changes post-harvest [4]. This variability in maturity can lead to increased risks of spoilage in overripe fruits, causing considerable losses during transport and storage [6]. Today's fruit monitoring methods, such as compression and puncture tests [7], are destructive and can reduce the profitability of the produce by damaging the goods during the assessment process. Existing

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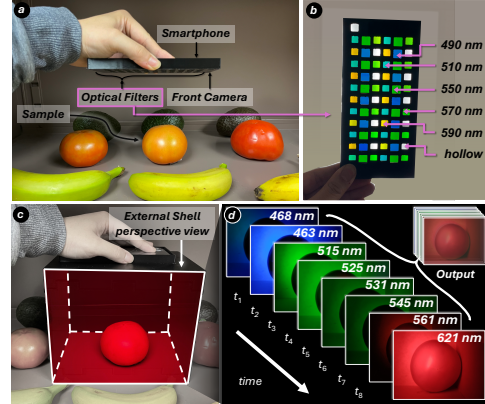


Figure 1: MobiChem Toolkit.

non-destructive techniques remain not deployable for end-user applications in the foreseeable future due to their complexity and high cost, especially for homes and grocery stores.

In this paper, we introduce MobiChem, a novel fruit monitoring system deployed on COTS smartphones. As shown in Figure 1, MobiChem captures images across numerous narrow, contiguous optical bands, allowing for a detailed analysis of fruits by identifying subtle physiological changes. Specifically, MobiChem reuses the phone screen to emit multiple narrow-band light spectrums toward the fruit. Using the front-facing camera, it captures the spectral profile of materials from reflected light, subsequently estimating fruit ripeness, chemical concentrations, and physical factors.

2 SYSTEM DESIGN

As shown in Figure 2, MobiChem consists of a *Smartphone-based Toolkit* hardware design and *Fruit Properties Analysis* algorithm.

The hardware module integrates a passive, foldable, and portable toolkit that can be easily attached to COTS smartphones for everyday use. The toolkit design module includes a phone-specific 3D printable outer shell and a custom backlight control application. First, geometric measurements, including phone dimensions, display size, camera position, and field of view, define the toolkit's outer shell and filter rack. Second, the optical characteristics of the smartphone's display are examined, focusing on its spectral response at full brightness across each color channel. By accurately quantifying this response, narrowband filters can be simulated to generate additional spectral channels. These optical measurements optimize filter placement for uniform illumination at each narrowband wavelength. During continuous image acquisition, the backlight control application sequentially activates and deactivates specific filter regions for desired wavelengths. Multispectral imaging is then achieved using the phone's front-facing (selfie) camera, with the synchronized activation and deactivation of specific filter regions on the display serving as the light source.

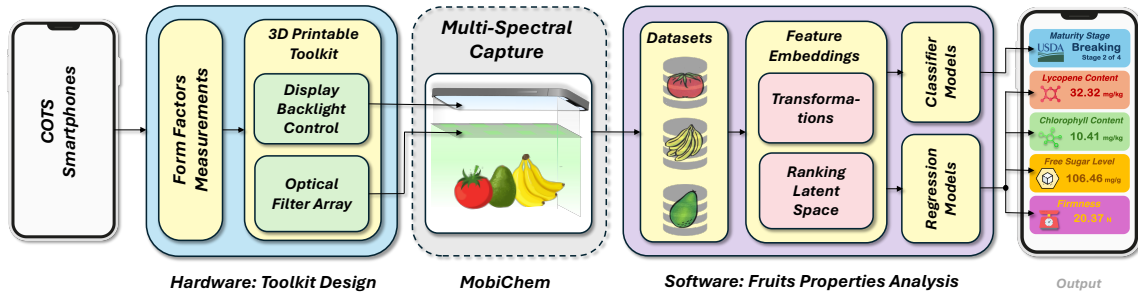


Figure 2: System Overview

MobiChem uses its raw HSI data for the application of fruit property analysis. To boost the accuracy of fruit stage recognition, a feature embedding module is proposed to enhance the model performance and reduce the bias. This module uses the Rank-N-Contrast to map the raw input into a continuous feature space.

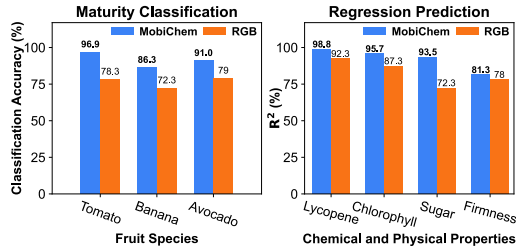


Figure 3: MobiChem vs RGB

3 IMPLEMENTATION

Data Collection Using Real Fruits. We monitor tomatoes, avocados, and bananas for our experimental analysis. We initiated the study by acquiring fruits of each type in unripe stage, and natural ripening was monitored closely. This process included 117 tomatoes across six maturity stages, 98 bananas across seven maturity stages, and 73 avocado across four maturity stages, with both RGB and MobiChem images captured.

Ground Truth. We used a KONICA MINOLTA CR-300 Chroma Meter [5], which has been widely used in previous studies [1, 2, 8] to correlate colorimeter readings with maturity grading, lycopene, chlorophyll, sugar content, and firmness in real fruits.

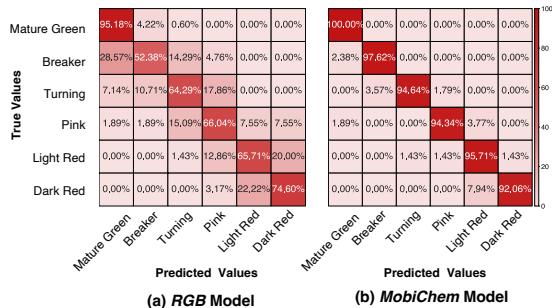


Figure 4: Confusion Matrix Classifying Tomato Maturity

4 EVALUATION

Maturity Stages Classification. The results in Figure 3 (left) clearly show the advantage of MobiChem for maturity classification.

MobiChem outperforms RGB by 18.6%, 14.0%, and 12.0%, respectively. The detailed confusion matrices in Figure 4 further illustrate the differences in stage-wise classification performance between RGB and MobiChem models for tomatoes. This indicates that MobiChem’s enhanced spectral resolution allows for detecting subtle variations in biological characteristics, helping to differentiate between ripening phases.

Chemical Concentration and Firmness Prediction: The results in Figure 3 (right) shows MobiChem demonstrates consistent gains over RGB across regression tasks, with improvements of 6.5% in lycopene, 8.4% in chlorophyll, 21.2% in banana sugar content, and 3.3% in avocado firmness.

5 CONCLUSION

In this paper, we presented MobiChem, a smartphone-based toolkit for practical fruit monitoring. Experiments on tomatoes, avocados, and bananas showed high accuracy: 95.67% for chlorophyll, 98.76% for lycopene, 93.53% for sugar, and 91.34% average for maturity classification. We believe such an application has the potential to transform daily consumer grocery habits and traditional agricultural practices, as well as enhance food assessment.

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